

Local Time Impact on Geomagnetic Storm response at mid-low latitudes

Castellanos-Velazco C. I.^{1,3} Corona-Romero P.^{2,3} Sergeeva M.^{2,3}
Gonzalez-Esparza J.A.

¹Universidad Nacional Autonoma de Mexico, Posgrado en Ciencias de la Tierra

² Laboratorio Nacional de Clima Espacial

³ Instituto de Geofisica

Abstract

The present study seeks to analyze longitudinally, the magnetic fluctuations associated with Ionospheric currents using 11 observatories encircling the globe at mid-low latitudes. The event to be studied is the geomagnetic storm known as the Saint Patrick event due to his main phase was developing during that celebration (march 17th, 2015). We perform Fourier analysis to reconstruct the magnetic signatures associated to these Ionospheric Currents. At the same time, through a wavelet spectrum in combination with a semblance analysis, we study the Disturbed Dynamo magnetic signatures. Results show that these Ionospheric magnetic signature responses are highly dependent on longitudinal location during The geomagnetic storm's 'Main Phase.

Keywords: Disturbed dynamo, wavelet

1 Introduction

A geomagnetic storm (GS) is a temporary disturbance of Earth's magnetosphere caused by the injection of energy and charged particles into the system, leading to a significant reduction in the geomagnetic field intensity. These storms are typically triggered by solar phenomena such as Coronal Mass Ejections (CMEs) or high-speed solar wind streams. Due to their potential impacts on technological systems—including disruptions to communications, navigation, and power grids—GSs are among the most extensively studied phenomena in space weather research.

During the Main Phase (MP) of a GS, energy and charged particles are transported into Earth's inner magnetosphere. Since the magnetosphere is dynamically coupled with the ionosphere, particularly at high latitudes [C.T Russell, 2016], the ionosphere exhibits pronounced responses to these perturbations. One critical ionospheric effect is the generation of storm-time electric fields, which manifest differently depending on latitude:

- Prompt Penetration Electric Fields (PPEFs) [Nishida, 1968a]
- Disturbed Dynamo Electric Fields (DDEFs) [Blanc and Richmond, 1980]

The concept of PPEF was first introduced by Nishida [1968b,a], who observed periodic magnetic fluctuations between 4 hours and 30 minutes in ground magnetometers located around the auroral oval and the dip equator, both showing nearly identical fluctuations. Later, Vasyliunas [1970], Jaggi and Wolf [1973] developed the first model explaining the penetration of convective electric fields ($\mathbf{E}_C$) during disturbed periods, propagating through Region 1 Field-Aligned Currents (R1-FAC). During quiet periods, $\mathbf{E}_C$ is shielded by a polarized electric field induced by Region 2 Field-Aligned Currents (R2-FAC). However, during the MP of a GS, magnetic reconnection leads to an almost immediate intensification of R1-FAC activity due to the injection of solar wind particles and energy. Since R2-FAC depends on ring current activity to strengthen, it initially fails the shielding of $\mathbf{E}_C$, allowing its transfer to the ionosphere through R1-FAC. This process continues until R2-FAC intensifies its activity through the intensification of ring current, allowing R2-FAC balancing R1-FAC activity, which can take from tens of minutes to several

hours [Wei et al., 2015]. During this period of imbalance, $\mathbf{E}_C$ penetrates into the mid- and low-latitude ionosphere, being that the reason for be called Penetration Electric fields. Furthermore, Kelley et al. [1979], Wei et al. [2015] showed that PPEF depends on the southward orientation of the interplanetary magnetic field (IMF) component B_Z and can also be triggered by sharp directional changes in B_Z .

DDEF is associated with an ionospheric current of charged particles injected into the auroral ionosphere, Joule heating occurs, leading to the circulation of equatorward thermospheric winds that drag charged particles (protons) Blanc and Richmond [1980]. When. This charge separation induces a Pedersen current in the same direction, and to the Coriolis effect, proton accumulation and westward circulation occur around the dip equator, generating a poleward electric field. Consequently, a second Pedersen current is induced in the poleward direction, opposing the first current. The above commented phenomena, at mid-latitudes ($\sim 40^\circ$), Hall currents are induced and then interrupted at their terminals and close at adjacent latitudes. This process results in vortex-shaped currents known as the Disturbed Dynamo, characterized by directions opposite to the Solar Quiet (SQ) current. For this reason, the disturbed dynamo is also referred to as the anti-SQ current.

Both PPEF and DDEF were studied by Fejer et al. [1983], who observed their effects using incoherent scatter radar at Jicamarca latitude. Later, Amory-Mazaudier [1982] detected DDEF above Saint-Santin latitude, with results consistent with the disturbed dynamo model. Later Le Huy and Amory-Mazaudier [2005], who modeled the magnetic field associated with ionospheric disturbances resulting from DDEF and PPEF contributions. To distinguish between both effects, Le Huy and Amory-Mazaudier [2005] considered that during the MP, the primary magnetic field contribution comes from PPEF, whereas during the recovery phase (RP), regional magnetic field variations are mainly driven by DDEF. Following this initial approach, several studies [Le Huy and Amory-Mazaudier, 2008, Zaka et al., 2009] have focused on analyzing local responses at different latitudes and longitudes to analyze the different local responses associate to DDEF phenomena. In Liu et al. [2013], a statistical study was conducted to understand ionospheric responses as a function of local sectors during GS onset. Later, Younas et al. [2020] proposed using Fourier analysis to isolate magnetic fluctuations related to DDEF and PPEF based on their quasi-periodic behavior, identifying periods of approximately 24 hours for DDEF and less than 4 hours for PPEF. Finally, Younas et al. [2021] combined cross-wavelet analysis [Torrence and Compo, 1998] with semblance analysis [Cooper and Cowan, 2008] to highlight DDEF magnetic signatures while filtering out other sources with similar fluctuation periods.

In this paper, we adapt the approach of Younas et al. [2021] to analyze DDEF and PPEF magnetic signatures at different longitudes during the geomagnetic storm (GS) known as the St. Patrick's event on March 17 – 18th. This event is recognized as the strongest GS of the 24th solar cycle. The selected ground magnetometers are located at 11 magnetic observatories distributed across different time zones while maintaining a similar magnetic latitude.

2 St. Patrick Geomagnetic Storm

[Introduction]

2.1 Interplanetary Precursor

From Figure 1, we present a brief description of the trigger event that led to the St. Patrick's GS. A more detailed description of the trigger event is provided by Wu et al. [2016].

Panel (a) shows the Interplanetary Magnetic Field B_T (black line) and its component B_Z (red line). Panels (b) and (c), on the other hand, present a few properties of the solar wind (SW), including its flow speed V_X (here taken as positive in the anti-sunward direction, black line), the proton temperature (blue line), and the proton density (orange line), as well as the Interplanetary Electric Field, which results from $V_X \times B_Z$. Finally, in panel (e), we present the planetary geomagnetic response, represented by the geomagnetic index SYM-H (green line).

The trigger event was highlighted using three vertical dashed lines within panels (a)–(d), as shown in Figure 1. The first dashed line (from left to right) indicates the arrival of the Interplanetary Shock (IPS) at 0430 UT, detected at L1. The IPS signature is clearly seen as a step-like effect in B_T and B_Z , as well as in the solar wind speed V_X and proton density n_P , while it is less noticeable in T_P . The arrival of the IPS in Earth’s near-space environment led to a compression effect on the magnetosphere, resulting in a temporary enhancement of Earth’s magnetic field. This enhancement is seen in the SYM-H index between 0430 and 0700 UT in panel (e), highlighted by light gray shading, and is known as Sudden Storm Commencement (SSC).

Later, the arrival of a CME’s sheath was identified by the highly fluctuating behavior of B_T and B_Z , as well as an enhancement in n_P and T , which are characteristic of a sheath region. It is important to note that the B_Z component changes direction southward between 0600 and 0900 UT, reaching minimum values around -18 nT. Consequently, there is a downward trend in the SYM-H index (panel e), marking the beginning of the main phase (MP) until 0900 UT, when the B_Z component shifts northward. The minimum SYM-H value reached during this period is approximately -80 nT.

Finally, we indicate the crossing of the CME near 1400 UT (marked by the second dashed line). As indicators, we highlight a less fluctuating behavior in B_T and B_Z , as well as a decrease in temperature in panel (d). With the arrival of the CME, a new period of southward B_Z begins, leading to a second phase of magnetic reconnection. Consequently, we observe a second decrease in the SYM-H index from 1200 UT until 2300 UT, reaching a minimum value of -228 nT. The entire MP period is highlighted by gray shading in panel (e). In panel (f), we highlight the period of interest with light gray shading, where we observe higher injection due to the MP.

3 Analysis

3.1 Geomagnetic Data

The geomagnetic data analyzed in this paper were downloaded from the INTERMAGNET public database (*International Real-Time Magnetic Observatory Network* [INTERMAGNET, 2021]), corresponding to the observatories (OB) listed in Table 1. Each OB is located within a narrow range of magnetic latitude ($21 - 33^\circ$). They are located also at different Universal Time (UT) zones.

Considering that some OBs are closer to others, the longitudinal study was divided into five zones based on the UT of each OB:

1. Zone 1: GUI (UT 0), TAM (UT 1)
2. Zone 2: JAI (UT 5:30)
3. Zone 3: LZH, BMT (UT 8), CYG, KAK (UT 9)
4. Zone 4: HON (UT -10)
5. Zone 5: TEO (UT -6), SJG (UT -4)

Although the distance between each OB presents a limitation for analyzing the longitudinal development of the GS, we can still examine the response within different time zones in which each OB was located during the MP.

3.2 Magnetic fluctuations of ionospheric origin

It is well accepted that [Rishbeth and Garriott, 1969, Knecht and B.M, 1985, Gjerloev, 2012, van de Kamp, 2013] local recordings H_{loc} of a ground magnetometer can be considered a linear combination of different geomagnetic sources:

$$H_{loc} = H_{SQ} + H_0 + H_{MT} + H_I, \quad (1)$$

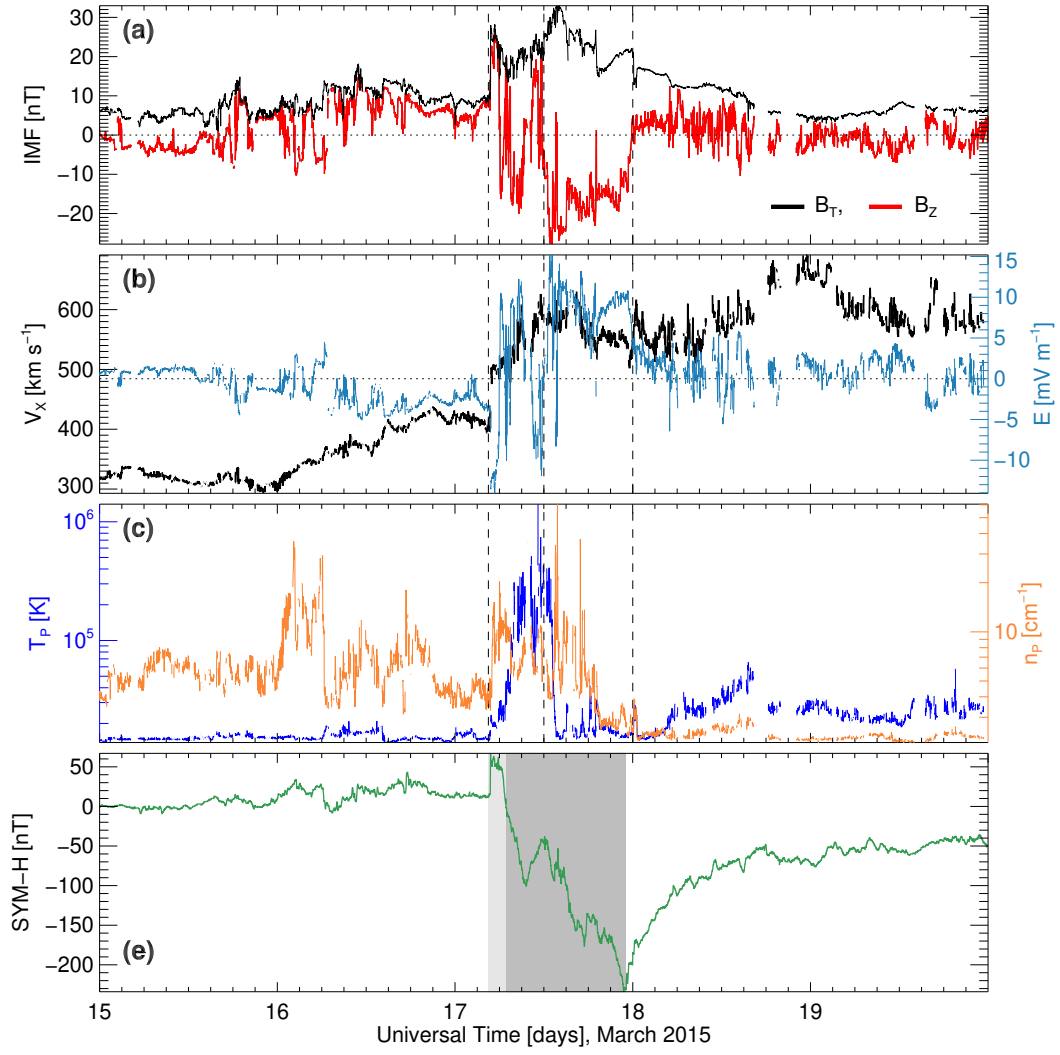


Figure 1: Interplanetary measurements. Panel (a): interplanetary magnetic field total component (black line) and z component (red line). Panel (b): V_X component of solar wind velocity (black line) and Interplanetary Electric Field (blue line). Panel (c): Proton Temperature (blue line) and Proton Density (orange line). Panel (d): SYM-H index (green line), highlighting the initial phase (light gray shade), main phase 1 (medium gray shade) and main phase 2 (dark gray shade).

Table 1: List of observatories considered for this study.

Country/ Territory	Observatory	IAGA code	geographic latitude	geographyc longitude	magnetic latitude	magnetic longitude	UT Zone h
China	Beijing	BMT	40.3 N	116.2 E	30.53 N	172 W	8
South Korea	Cheongyang	CYG	36.37 N	126.854 E	27 N	162.43 W	9
India	Jaipur	JAI	26.917 N	75.8 E	18.42 N	150.47 E	5.5
Spain	Guimar	GUI	28.317 N	16.433 W	33.21 N	61.1 E	0
USA	Honolulu	HON	21.317 N	158 W	21.49 N	90.03 W	-10
Japan	Kakioka	KAK	36.232 N	140.186 E	27.8 N	149.73 W	9
China	Lanzhou	LZH	36.083 N	103.833 E	26.23 N	176.79 E	8
USA, Puerto Rico	San Juan	SJG	18.11 N	66.15 W	28.19 N	6.1 E	-4
Argelia	Tamanraset	TAM	22.783 N	5.571 E	24.3 N	82.29 E	1
Mexico	Teoloyucan	TEO	19.747 N	99.182 W	27.81 N	28.4 W	-6

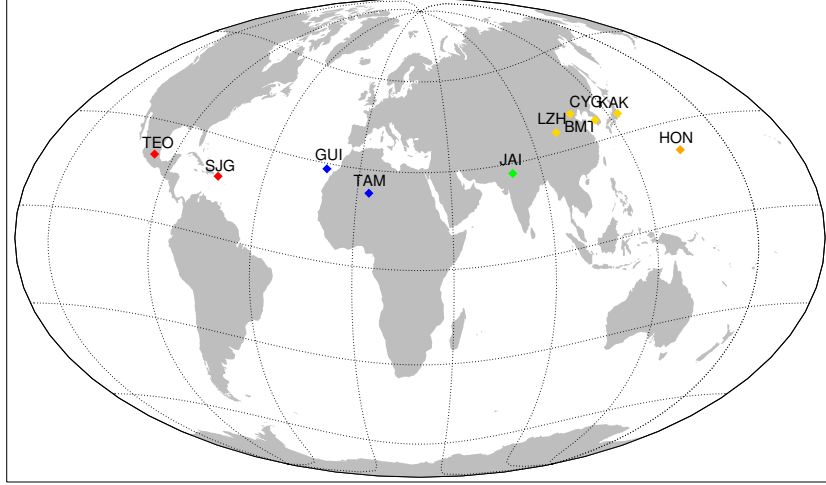


Figure 2: Geographic position of every magnetic OB considered for this study. The color code corresponds to different local time regions: blue (UT 0, 1), green (UT 5:30), yellow (UT 8, 9), orange (UT -10), red (-6, -4).

where H_{SQ} represents the diurnal variation and H_0 is the main field during a month window. H_{MT} represents the variations induced by the intensification of magnetospheric currents during the GS. Finally, H_I corresponds to the magnetic variations driven by ionospheric current activity during a GS. Thus, the local geomagnetic response during a GS is given by:

$$\Delta H = H_{loc} - (H_{SQ} + H_0). \quad (2)$$

We provide a detailed explanation of how we model the regular variations in Appendix A.

From Le Huy and Amory-Mazaudier [2005], Amory-Mazaudier et al. [2017], Younas et al. [2020], for mid-low magnetic latitudes, with λ as the geomagnetic latitude of each OB, we can approximate H_{MT} as:

$$H_{MT} \approx SYMH \cdot \cos(\lambda), \quad (3)$$

Thus, H_I can be approximated as:

$$H_I = H_{DDEF} + H_{PPEF} \approx H - (H_{SQ} + H_0 + SYM - H \cdot \cos(\lambda)), \quad (4)$$

where H_{DDEF} and H_{PPEF} represent the magnetic fields associated with the disturbed dynamo electric fields and the prompt penetration electric fields, respectively.

In order to isolate H_{DDEF} and H_{PPEF} from each other, we proceed under the assumption that both ionospheric currents drive quasi-periodic fluctuations [Nishida, 1968b, Blanc and Richmond, 1980, Younas et al., 2020]. Hence, we begin with a Fourier analysis, applying passband filters to extract fluctuations with frequencies around $11.5\mu\text{Hz}$ (24-hour periods) to reconstruct H_{DDEF} . For H_{PPEF} , we apply high-pass filters for frequencies higher than $69.5\mu\text{Hz}$ (corresponding to periods ≤ 4 hours).

Additionally, with the purpose of analyze the time domain, we apply wavelet techniques [Torrence and Compo, 1998], following Younas et al. [2021], where they compare two time series (H_{SQ} and H_I) with similar band frequency fluctuations. The goal is to identify which signal is more anti-correlated with H_{SQ} and, consequently, more correlated with H_{DDEF} activity.

More specifically, we perform a wavelet analysis (W), computing the wavelet spectrum ($|\psi_0|^2$) to highlight the power of the signal related to H_I and the cross-wavelet transform (XWT) [Torrence and Compo, 1998]:

$$W_n^{SQ, H_I} = W_N^{H_{SQ}} W_N^{H_I*}, \quad (5)$$

where N is the index of each time series, W_n^{HSQ} represents the wavelet transform applied to the diurnal variation, while W_n^{HI*} is the conjugate of the wavelet transform applied to H_I .

The local phase θ , an output from 5, can be used to compare two datasets based on correlations between their phase angles (θ) as a function of frequency meanwhile α (another output from XWT analysis) is used as scale factor. Then, the coherence analysis is represented as:

$$S = \alpha \cdot \cos^n(\theta). \quad (6)$$

It is important to note that the result of Equation 6 includes not only the H_{DDEF} signal but also the signals related to SQ and PRC [Younas et al., 2020, 2021]. To completely isolate H_{DDEF} in the spectrogram, we normalize the result of Equation 6 to a range of $[-1, 1]$, as it is easier to separate normalized signals. However, in this case, we reverse the scale so that values closer to 1 can be interpreted as H_{DDEF} . Since values closer to 0 are not related to either H_{SQ} or H_{DDEF} , we consider correlations greater than 0.5 as indicative of H_{DDEF} , as shown below:

$$\alpha \cdot S = \begin{cases} 0, & \text{if } S_i < 0.6, \\ H_{DDEF}, & \text{if } S_i \geq 0.6. \end{cases} \quad (7)$$

Having the wavelet spectrum $|\psi_0(n)|^2$, which gives us the power of the signals, and $\alpha \cdot \text{anticorr}(\text{semblance})$, which highlights the signal related to H_{DDEF} , we can observe not only whether H_{DDEF} variations are present during the event but also their intensity.

4 Results

In the following section, we present the geomagnetic response to the St. Patrick's GS for each OB listed in Table 1. Then we show the results of Equation 4 for isolating H_I , as well as the reconstruction of H_{DDEF} and H_{PPEF} through Fourier analysis. Finally, we provide the results of Equation 6 to isolate the H_{DDEF} signal in the time-frequency domain.

4.1 Geomagnetic and Ionospheric response

In each panels of Figure 3, we show the global indices SYM-H and ASYMH, which represent a quantitative measure of the ring current and partial ring current activity, respectively. In bottom panels, we show the results of Equation 4 (red line) in comparison to H_{SQ} modeled for each OB (see Appendix A for more details). Within each panel, we highlight local nighttime periods with light gray shading, set from 1800 to 0600 LT each day. This makes it easier to indicate differences in LT for each OB during and after the MP.

Beginning with **Zone 1** (GUI and TAM plots) at UT-0 where the first stage of the MP (triggered by the CME's sheath) developed on day side. The second stage of the MP (triggered by the CME) occurred partially on the day side and night side, almost centered on the dusk sector (1800 LT). Both cases show an enhancement of ΔH near local dusk, which is also consistent with an enhancement in ASYMH activity, especially in TAM's local response. In GUI, ΔH reaches its minimum near 1900 LT, while TAM's minimum occurs at 1800 LT. After their minima, we observe that the local magnetic response remains enhanced until the minimum at SYM-H, which coincides with a second period of ASYMH activity near 2300 LT.

In bottom panels, there is sharp crests related to H_{SQ} activity near noon, while $H_{I(GUI,TAM)}$ presents two responses. The first, the most intense response, is developed during the MP and can be related to both PRC activity and H_{PPEF} fluctuations, with peak amplitudes of approximately -110 nT (GUI) and -150 nT (TAM). These responses are mostly observed around the dusk sector, similar to ASYMH activity, with TAM showing more intense activity. The second effect of interest is the fluctuations associated with H_{DDEF} . These fluctuations appear several hours to a day after the MP's conclusion, as mentioned by Le Huy and Amory-Mazaudier [2005]. We can only consider H_{DDEF} fluctuations when H_I is at least one day after the ΔH minima. Another condition we could appoint is that H_I fluctuations have to be approximately opposed to H_{SQ} . In other words, whenever there is a maximum (minimum)

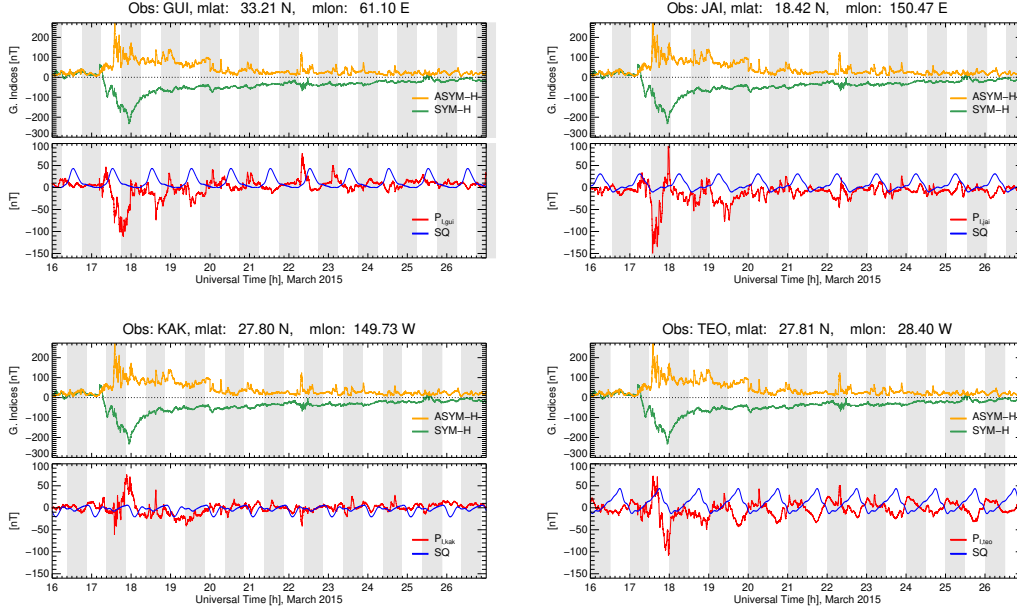


Figure 3: Geomagnetic observations for each magnetic OB. top panels show planetary geomagnetic indices SYMH (green line) and ΔH (black line). Bottom panels: Magnetic contribution of ionospheric origin (red line) and diurnal variation modeled from local quiet days (blue line) (Part 1)

in H_{SQ} , there is a minimum (maximum) in H_I . Therefore, $H_I = H_{DDEF}$ between March 18th and 21st.

For JAI in **Zone 2**, it is noticeable that MP on the day side exhibits similar responses for both the global and local magnetic fields. After reaching the dusk sector, the local response shows a sharp enhancement compared with SYMH; where ΔH reaches its minima at 2100 LT. Similar to GUI and TAM, this enhancement is consistent with the sharp increase in ASYMH.

Zone 3 OBs have in common that second stage of MP developed during local night time, reaching a minima ΔH after the midnight sector. Top panels of LZH, BMT, CYG, and KAK, show enhanced ΔH activity compared to SYMH, coinciding with ASYMH highest activity. However, these enhancements are more attenuated as we move eastward, although some fluctuations are still present around the midnight sector. Once the OBs of **Zone 3** are near the dawn sector, there is a sharp increase in ΔH , which is not present in SYMH.

In bottom panels, there is something interesting about both H_{SQ} and H_I behavior. In the case of regular variation, instead of a crest, there is a valley around noon. On the other hand, H_I shows a decrease in the negative enhancements during ASYMH main activity (a few hours before each local midnight), as we move eastward, which is consistent with the decrease in ΔH shown in the top panels. The positive enhancement around the dawn sector, however, increases as we move eastward. In the case when $H_I \approx H_{DDEF}$, we do not observe remarkable fluctuations, which could indicate that there was no presence of H_{DDEF} in **Zone 3**.

In **Zone 4**, represented by HON, the MP phase develops between 2100 and 1400 LT, with the period triggered by the CME occurring during the day side. Unlike the previous cases, ΔH shows less negative values compared to SYM-H during the highest activity of ASYMH. This happens when HON is near the dawn sector. Near the noon sector, the minimum of SYM-H behaves similarly to ΔH , with about a 1-hour difference between each minimum.

H_{SQ} in the bottom panel shows again a sharp crest pattern around noon. H_I shows a positive enhancement during the peak of ASYMH activity, consistent with the behavior observed in the top panel between ΔH and SYM-H. As for H_{DDEF} , some possible fluctuations are observed between March 19th and March 21st.

Finally, in **Zone 5** (TEO and SJG), MP (triggered by CME) occurs mostly during day side. On one hand, there is a positive enhancement of ΔH during the peak ASYMH activity, locally around 800 LT (TEO) and 1000 LT (SJG). On the other hand, the minima of ΔH is consistent in time with SYM-H, but it shows more negative values. In other words, the local response surpasses the global response. In both cases minimum value is reached after TEO and SJG crossed dusk sector (at 1800 and 2000 LT respectively).

Bottom panels show a first positive H_I enhancement of 70 nT in TEO between 0700 and 1000 LT, and 50 nT between 0900 and 1200 LT in SJG. This is followed by enhanced negative activity with amplitudes of -108 and -99 nT, both happening around dusk (1800 LT). While SJG shows $H_I \approx H_{DDEF}$ activity during March 19th to 21st, in TEO, $H_I \approx H_{DDEF}$ fluctuations remain visible until March 26th, making it the OB where H_{DDEF} seems to last the longest, with a very smooth decay in amplitude.

4.2 Reconstruction of H_{PPEF} and H_{DDEF}

The results of applying frequency band filters to the FFT(H_I) are shown in Figure 4. Fluctuations related to H_{PPEF} are represented by red lines while H_{DDEF} fluctuations are represented by black lines. Similar to Figure 3, each local night time period is highlighted with gray shading.

In **Zone 1**, H_{PPEF} fluctuations begin with the onset of the MP. However, the Sheath triggered MP stage did not drive significant H_{PPEF} fluctuations compared to the CME. After the arrival of the CME, there is an enhancement tendency from noon sector toward dusk sector. In both GUI and TAM, there is a sharp fluctuation of -30 nT and -45 nT. Following this, a decreasing trend in amplitudes begins until midnight, when a new sharp peak in H_{PPEF} is observed, reaching 34 nT for GUI and 60 nT for TAM. This enhancement is consistent with the minimum in SYM-H and the end of magnetic reconnection mechanism, as seen in Figure 1.

Based on the previous Figure 3 and according to Le Huy and Amory-Mazaudier [2005], we consider that H_{DDEF} become valid at least one day after the minimum. Then, from March 19th at midnight is when we can consider H_{DDEF} as valid. Initially, in both GUI and TAM, the maximum amplitude is -20 nT, but these amplitudes decrease rapidly until March 21st. In the case of GUI, however, there is a small enhancement between March 22nd and March 23rd.

Fluctuations H_{PPEF} in JAI (**Zone 2**) show a tendency during MP related to the CME at night hours. Centered around midnight, appear the highest amplitudes above 40 nT meanwhile the maximum amplitude is 52 nT, near dawn sector. An enhancement of H_{PPEF} is also present near local dusk, with a magnitude of -41 nT. H_{DDEF} , on the other hand, presents a similar behavior as in Zone 1, with an initial amplitude of about 20 nT on March 19th, decreasing until March 21st.

In **Zone 3**, similar to the previous case, most of the relevant H_{PPEF} fluctuations occur during night hours, including the first two hours after dawn. The H_{PPEF} fluctuations driven by the CME show a sharp enhancement close to each local midnight within this zone. The maximum fluctuation is observed around each local midnight, with values of 50, 55, 47, and 30 nT for LZH, BMT, CYG, and KAK, respectively. After each local dawn, there is another enhancement of H_{PPEF} , followed by a sharp decrease in amplitudes. On the other hand, H_{DDEF} fluctuations are practically absent in OBs within this zone, with an initial amplitude of about 10 nT at most, starting from March 19th.

In HON (**Zone 4**), the H_{PPEF} fluctuation tendency has considerably smaller amplitudes compared to the previous cases. The maximum fluctuations occur at 2300 LT and 0700 LT, with magnitudes of 34 and 41 nT, respectively. Considering that H_{DDEF} would be valid until March 19th-20th for this zone, there are no significant H_{DDEF} fluctuations present in HON either.

Finally, for **Zone 5**, we observe that H_{PPEF} fluctuations related to the CME occur during daylight hours. The amplitude tendencies exhibit a very smooth variation in TEO and SJG, with the peak at noon for TEO, while for SJG, it occurs close to 1400 LT, with significantly weaker fluctuations. In both cases, the maximum H_{PPEF} fluctuations occur in two moments: the first at noon (TEO) or 1400 LT (SJG), and the second at dusk. In the case of TEO, the maximum H_{PPEF} fluctuations are about 34

nT, while SJG's maximum fluctuations are about 25 nT.

H_{DDEF} fluctuations shows a quite interesting behavior. In the case of TEO, H_{DDEF} presents a higher amplitude of 20 nT compared to HON. Its amplitude damping is also the slowest within this study, making TEO the OB where H_{DDEF} lasts the longest, from March 19th to March 26th. In the case of SJG, H_{DDEF} shows a different behavior, being more similar to what is observed in HON.

It should be considered that the time at which H_{DDEF} becomes valid in all cases is only an approximation. To assess the period during which H_{DDEF} was present, as well as its intensity, the next procedure demonstrates the results.

4.3 Wavelet Analysis

Results of the wavelet analysis for each OB are shown in Figure ?? . Panels a) display local response ΔH as a reference for the GS development. Panels b), on the other hand, show $|\psi_0^2|$ to analyze the power of the signals at $f \approx 11.5\mu\text{Hz}$ or $T \approx 24$ hours, as well as when these signals were present. As shown in Figure 4, during the MP of every OB, a peak signal is observed which is not related to H_{DDEF} since it takes several hours to a whole day after minimum is reached, for the Ddyn current to be driven. Thus, we use panel c) to display the normalized semblance analysis (S), where we compare two signals with similar frequency bands (P_I and SQ). Here, the values range from 0 to 1, where values from 0.6 to 1 (green to red) are more anti-correlated to SQ and therefore, more correlated to H_{DDEF} . By examining all three panels, we can study the H_{DDEF} local response for each OB.

According to the wavelet spectrogram and semblance analysis in Figure ??, for the **Zone 1** OBs, H_{DDEF} is present since March 19th (around 0600 LT in GUI and 1900 LT in TAM) and remains until almost the end of March 21st. As seen in Figure 4, in GUI, there is a minor intensification of H_{DDEF} . The power of the signal in GUI at the beginning is about $4.25 \times 10^4 \text{ nT}^2/\text{Hz}$, with a damping of $1.17 \times 10^4 \text{ nT}^2/\text{Hz}$ by March 21st. This power remains until March 24th, when the signal's correlation drops to 0.5. In the case of TAM, the maximum amplitude during the first day is about $2.55 \times 10^4 \text{ nT}^2/\text{Hz}$, and it drops to $1.17 \times 10^4 \text{ nT}^2/\text{Hz}$.

In **Zone 2**, the wavelet spectrogram shows a similar response to TAM. However, the coherent analysis reveals a different result for JAI. The background shows a correlation around 0.5, while we observe notable signals within the frequency range of $13.8\mu\text{Hz}$ ($T \approx 20$ hours), with correlations ranging from 0.75 to 1 during March 19th and between March 25th and 27th. The first signal, which can be associated with H_{DDEF} , shows a power of 5.95×10^4 to $2.55 \times 10^4 \text{ nT}^2/\text{Hz}$, while the second period has a significantly weaker power of $8.5 \times 10^3 \text{ nT}^2/\text{Hz}$.

In **Zone 3**, there is no fluctuations related to H_{DDEF} in any of the OBs. The same applies to **Zone 4**, represented by HON, where the signal identified as anticorrelated with H_{SQ} appears at a frequency range of $23.1\mu\text{Hz}$ (period ≈ 12 hours), which is about twice the frequency range in which H_{DDEF} is present. A similar pattern is present in **Zone 3**, but the signals appear at even higher frequencies, ranging from 55.5 to $69.4\mu\text{Hz}$ (periods of $\approx 5 - 4$ hours).

Zone 5 exhibits quite different behavior. On one hand, we have TEO, where H_{DDEF} fluctuation power is approximately $4.4 \times 10^3 \text{ nT}^2/\text{Hz}$, and this power is maintained for almost six days starting from March 20th, the date when, according to the semblance analysis, H_{DDEF} begins its activity. Then, we observe a smooth attenuation of power to approximately $2.55 \times 10^4 \text{ nT}^2/\text{Hz}$ until March 26th, when the correlation drops below 0.5, marking the end of H_{DDEF} activity. Thus, TEO is the point where H_{DDEF} lasts the longest in this study.

On the other hand, SJG shows a similar response in the spectrogram to TAM's case, with H_{DDEF} beginning its activity on March 22nd and lasting for about three days. The power of the signal is approximately $8.5 \times 10^3 \text{ nT}^2/\text{Hz}$, which is one of the weakest observed in this study.

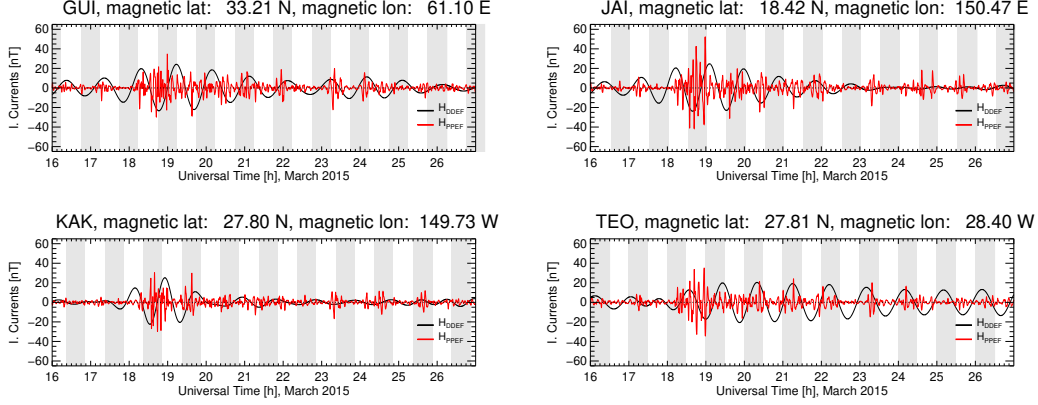


Figure 4: Geomagnetic registers for each magnetic OB. top panels show planetary geomagnetic indices SYM-H (green line) and local processed magnetic registers (black line). Bottom panels: Magnetic contribution of ionospheric origin (red line) and diurnal variation modeled from local quiet days (blue line) (Part 2)

5 Discussion

From Figures 3, 4, and ??, there are consistent and similar results for the OBs in each Zone. In **Zone 1**, the MP phase occurs mostly during day time hours, except for the last six hours. As shown in Figure 3, around 1400 UT (1500 LT in the case of TAM), there is an enhancement in the local magnetic response, which coincides with the enhancement of ASYMH. Since ASYMH is a good measure of the asymmetric response of the ring current system, it also reflects the contribution from the PRC. The top panels of Figure 3 show that the PRC influences local geomagnetic activity between 1400 UT and 2300 UT.

Considering the shape of the PRC, its contribution to ΔH is not uniform across the OBs, as observed in Figure 3. As we move eastward through **Zone 2**, **Zone 3**, and **Zone 4**, the influence of the PRC becomes less significant as the OB is further from the region of PRC activity. In **Zone 5**, on the other hand, the OB show less negative ΔH during the peak PRC activity. This is particularly noticeable when an OB cross the dawn sector during the MP. In these cases, there is an enhancement in ΔH , which could be related to the direction of electric currents and fields in this sector, driving a positive H component. Conversely, in the dusk sector, the direction of electric fields and currents may induce a negative H . Therefore, OBs near dusk sector during MP would present a stronger local response, while those near the dawn sector would show a more attenuated response.

The difference in local enhancement (either negative or positive) due to PRC activity for each OB influences the results of Equation 4. As a result, H_I exhibits different amplitudes and behaviors depending on the Zone. In **Zone 1**, H_I shows high negative amplitudes, corresponding to a more negative local response compared to the global one. On the other hand, from **Zone 2** to **Zone 4**, there is an attenuation of H_I between 1400 - 1900 UT. The reason for this attenuation is that as an OB gets farther from the PRC's extreme terminal, it results in less contribution from it. Additionally, we observe an enhancement of H_I toward more positive values between 2100 - 0300 LT, which corresponds to a more attenuated local response in comparison to SYM-H. In contrast, OBs in **Zone 5** show an enhancement of more negative values near local dusk, when SYM-H reaches its minimum. This suggests that OBS within **Zone 5** experience a sharp negative enhancement of PRC at the end of the MP.

As observed in Figure 4, H_{PPEF} peak fluctuations are higher in TAM and JAI. This is also evident in Figure 3, where ΔH_{loc} presents a higher enhancement from PRC contribution. On the other hand, OBs from **Zone 3** exhibit higher fluctuation tendencies in H_{PPEF} .

To quantify amplitude tendencies in general, we measured the distribution of H_{PPEF} during MP, as shown in Figure 5. From these distributions, we take $|H_{PPEF}|$ since we are primarily interested in their amplitudes. We then set thresholds for each OB, where any fluctuation exceeding the threshold is considered highly significant. These thresholds are computed by taking twice the standard deviation

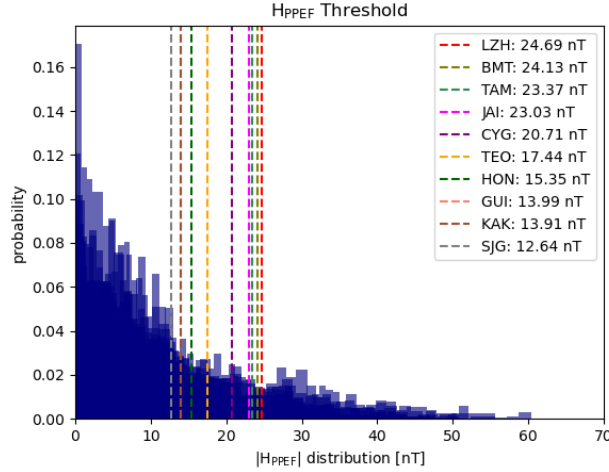


Figure 5: H_{PPEF} distribution for each OB. Color dashed vertical lines represent the a threshold for H_{PPEF} to be considered as intense amplitudes

for each distribution and are represented as dashed lines with different colors for each OB in Figure 5. It is evident that the highest thresholds correspond to OBs crossing between the dusk-midnight sector during the period of highest ASYMH activity (1325 - 1750 UT). These include LZH (2125 - 0150 LT), TAM (1425 - 1850 LT), JAI (1855 - 2320 LT), and CYG (2225 - 0150 LT). As observed, this indicates that PRC activity had a significant impact on the intensity of H_{PPEF} .

From Younas et al. [2020], it is clear that the PRC makes an important contribution to the local response, which could affect the results of Equation 4 and the reconstruction of H_{DDEF} . Considering the PRC's limited longitudinal region of influence, we decided to analyze its contribution to each OB.

From analyzing H_{PPEF} thresholds, we acknowledge the importance of focusing on the time window from 1325 to 1750 UT, during which PRC peak activity occurs. Therefore, by considering H_I and ASYMH from this time window, we computed a Pearson correlation (ρ) in two stages: first, ρ_1 for ASYMH - H_I , and second, ρ_2 for ASYMH - H_I after removing H_{PPEF} from H_I using a low-pass filter. The results are shown in Figure 6, where red diamonds represent ρ_1 , and gray diamonds represent ρ_2 .

Note that a few OBs exhibit a strong correlation (or anti-correlation), such as JAI, LZH, and TEO. Interestingly, at the beginning of this time window, JAI and TEO were also the closest OBs to the dusk and dawn sectors, respectively. On the other hand, after removing H_{PPEF} fluctuations from H_I , the scattering increases in all OBs, especially those with the highest ρ_1 . This reinforces the idea that H_{PPEF} amplitudes are strongly influenced by PRC activity.

To conclude the discussion on H_{PPEF} , there is another notable feature. At 1730 UT, a substorm onset is reported by Guineva et al. [2019], Guerrero et al. [2017]. From Hui et al. [2017], it is concluded that a substorm may affect the penetration efficiency of electric fields, even at mid-low latitude regions. Therefore, we infer that this substorm might influence H_{PPEF} at OBs located near the dusk, dawn, or midnight sectors between 1730 and 1800 UT.

Regarding H_{DDEF} fluctuations, results show distinct differences when we focus on more eastern OBs, starting with GUI. In **Zones 1 and 2**, fluctuations of H_I are present during the first three days after March 18th, whereas in **Zones 3 and 4**, these fluctuations are absent. This was confirmed through the reconstruction of H_{DDEF} and coherence analysis, which revealed no fluctuations related to H_{DDEF} in **Zones 3 and 4**. Notably, the gray shades in Figures 3 and 4, which represent local nighttime, show that the MP in **Zones 3 and 4** occurs during night hours. This aligns with the Ddyn model, which predicts that the current system is primarily driven on Earth's day side, although it can also be present on the night side, but at much lower intensity.

After the semblance analysis, we observe that in most OBs (except those in **Zones 3 and 4**), H_{DDEF}

lasts for a maximum of three days. The exceptions are GUI, which shows a minor intensification after its depletion, and TEO, where the amplitude drop is much slower, allowing H_{DDEF} fluctuations to persist for almost a week. The case of TEO is particularly noteworthy, as the nearest OB, SJG, does not exhibit a similar response.

In Wu et al. [2016], Younas et al. [2021], it is mentioned the arrival of a High Speed Solar Wind flow after march 18th. This can be seen in panel (b) of Figure 1 where SW speed remains above 550 kms^{-1} . However, looking at B_Z and E_Y profiles, there were no significant disturbing process but really small injection periods. This could explain the long duration of RP and so lasting effects of DDEF as mentioned in Younas et al. [2021]. According to them, whether the trigger event is a CME or High-Speed Solar Wind (HSSW), both the duration and longitudinal presence of Ddyn are affected. Their results indicate that a CME-triggered event may not be observed at all longitudes, as was the case here. In contrast, an HSSW-triggered event, even with less energy input than a CME, leads to short injections over a wider period, resulting in Ddyn lasting several days. The duration of Ddyn in TEO and GUI aligns with their findings, where Ddyn lasted about 4-5 days in the American and African sectors, and 1.5 days in Asia.

6 conclusions

The aim of this project was to analyze longitudinal magnetic fluctuations driven by ionospheric currents during an intense geomagnetic storm, such as the St. Patrick's event on March 17th, 2015. To achieve this, we used geomagnetic data from ground magnetometer OBs located across different time sectors.

Throughout this study, we analyzed the magnetic signatures associated with Prompt Penetration Electric Fields (PPEF) and Disturbed Dynamo Electric Fields (DDEF). We conducted a Fourier analysis to reconstruct H_{PPEF} and H_{DDEF} using high-pass and pass-band filters, respectively. Additionally, we applied the wavelet technique to isolate the H_{DDEF} signal from other sources within the frequency band of 11.5 μHz (period ≈ 24 hours).

After reconstructing H_{PPEF} for each OB, we reached the following conclusions:

- H_{PPEF} exhibits higher fluctuations on the night side, particularly when an OB crosses the dusk-midnight sector during peak PRC activity.
- This behavior is consistent with the correlation analysis, where the highest correlation between ASYM-H and H_I was observed for the same OBs that showed higher H_{PPEF} thresholds.
- The amplitudes of H_{PPEF} are significantly enhanced by PRC activity, especially when the OBs are near the dusk or dawn sectors.
- Some of the highest H_{PPEF} amplitudes occur around the time window during which two substorm onsets were reported (at 17:29 and 22:50 UT) by Guineva et al. [2019], Guerrero et al. [2017]. This suggests that substorms may also influence H_{PPEF} amplitudes for OBs near the dusk, midnight, or dawn sectors, which aligns with findings from Hui et al. [2017].

Regarding H_{DDEF} computed through both Fourier and Semblance analysis, we conclude the following:

- TEO and GUI (which show a brief re-intensification) are the OBs predominantly located on the day side during the magnetic storm. The duration of H_{DDEF} activity at these OBs is consistent with the results reported by Younas et al. [2021] for the same event but different OBs.
- In contrast, the OBs in **Zone 3 and 4** exhibit virtually no H_{DDEF} activity. This is in line with the disturbed dynamo model Blanc and Richmond [1980], which suggests that the drop in conductivity during the night side, along with the vanishing of the Ionospheric E region, leads to significantly weaker DDEF activity, and therefore weaker H_{DDEF} .

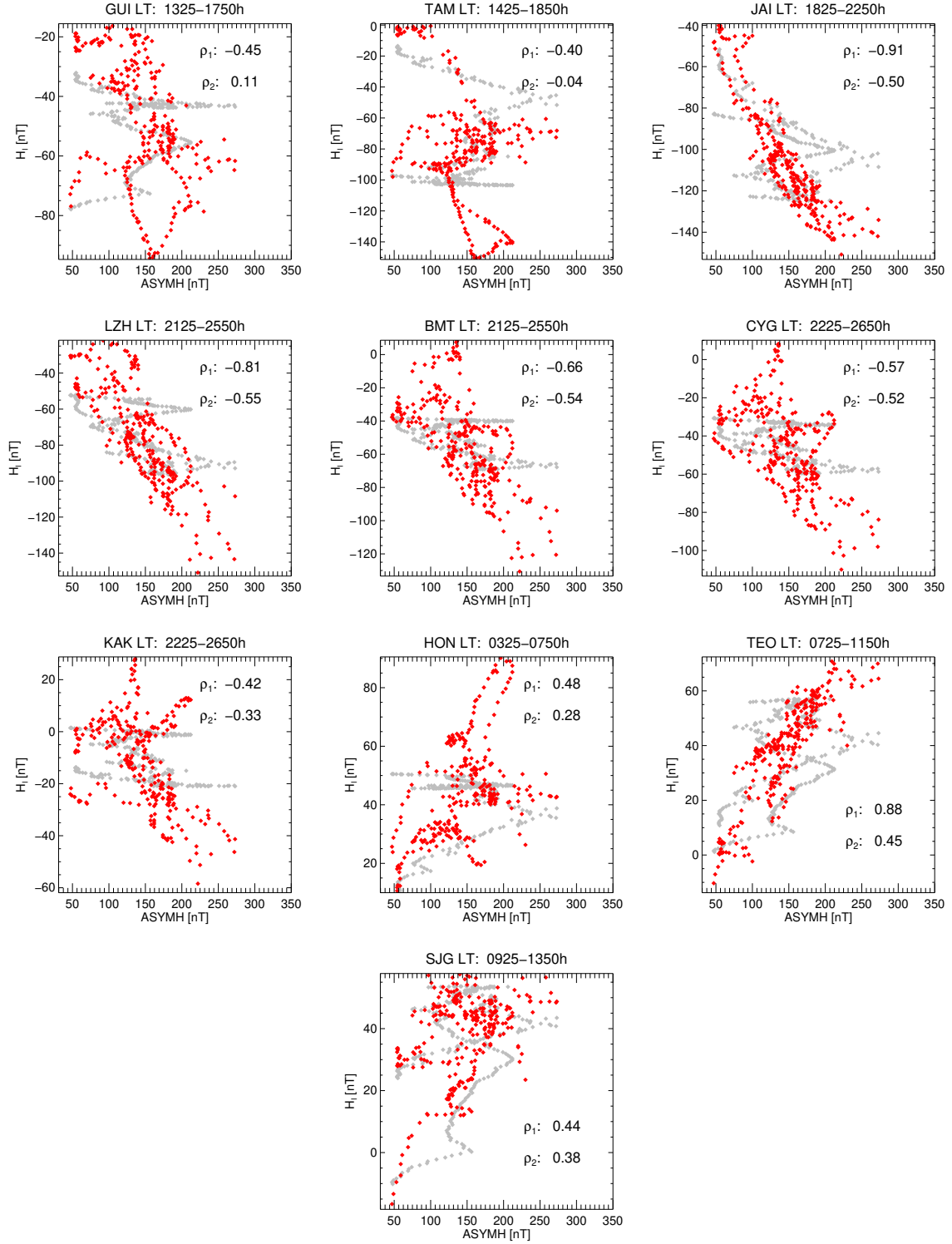


Figure 6: Correlation Analysis ASYM-H and H_I , during the period 13:25 - 17:50 UT, which is consistent with the period of highest ASYM-H activity

- The absence of H_{DDEF} magnetic effects in certain longitudes (**Zones 3 and 4**) is also consistent with Younas et al. [2021], where it was noted that a CME-triggered event might not be observed at all longitudes, even if the energy input is larger. The shorter duration of energy input in CME events compared to those triggered by high-speed solar wind (HSSW) explains the limited H_{DDEF} in these zones.
- In summary, OBs located on the night side during the main phase (MP) of the geomagnetic storm exhibit lower or even null H_{DDEF} activity. Conversely, the same event may produce a strong and prolonged H_{DDEF} if the OB is situated on the Earth’s day side.

A Processing magnetic data

It is crucial to eliminate the regular variations present in the magnetic records to isolate the magnetic contributions of interest. The main field H_0 , which can be removed by modeling a daily or monthly baseline, was approximated by computing either the daily mode or the center of a Gaussian fit [Gjerloev, 2012] for the local-night magnetic data $H_0 : 3$. Subsequently, H_0 was computed from the median of the daily values in March. To exclude daily values during disturbed days, we adapted the procedure outlined in van de Kamp [2013]. Specifically, we applied a moving window, calculating the hourly Interquartile Range (IQR). Disturbed days were identified by setting a threshold for variation, where:

$$IQR_{med}^i > threshold = disturbed \quad (8)$$

This threshold was set based on the distribution of the IQR peaks, where the maximum IQR (IQR_{max}) was selected by resampling the data using tri-hourly bins.

To compute the diurnal variation, we identified what we refer to as local quiet days (LQD) for each station. Following the methodology in van de Kamp [2013], we considered the five days with the smallest daily IQR peaks as LQDs. We then computed the average of the H component data during these five quietest days:

$$\langle LDQ_H(t) \rangle = \frac{1}{5} \sum_{i=1}^5 H_i(t) - \langle H_i 0 : 3 \rangle \quad (9)$$

where we normalize the offset of each daily data by subtracting the average of the first four hours of magnetic data in local time, as indicated by the second term in Equation 9. Subsequently, we model the diurnal variation H_{SQ} by applying a low-pass filter to the Fourier Transform of the result from Equation 9, following the procedure in Equation (1) of van de Kamp [2013].

B Other Events

Acknowledgements

References

- C. Amory-Mazaudier, O. S. Bolaji, and V. Doumbia. On the historical origins of the CEJ, DP2, and Ddyn current systems and their roles in the predictions of ionospheric responses to geomagnetic storms at equatorial latitudes. *Journal of Geophysical Research (Space Physics)*, 122(7):7827–7833, July 2017. doi:10.1002/2017JA024132.
- Christine Amory-Mazaudier. Electric currents above saint-santin. i - data. *Journal of Geophysical Research*, 87:2459–2464, 04 1982. doi:10.1029/JA087iA04p02459.
- M. Blanc and A. D. Richmond. The ionospheric disturbance dynamo. *Journal of Geophysical Research*, 85(A4):1669–1686, April 1980. doi:10.1029/JA085iA04p01669.
- G. R. J. Cooper and D. R. Cowan. Comparing time series using wavelet-based semblance analysis. *Computers and Geosciences*, 34(2):95–102, February 2008. doi:10.1016/j.cageo.2007.03.009.

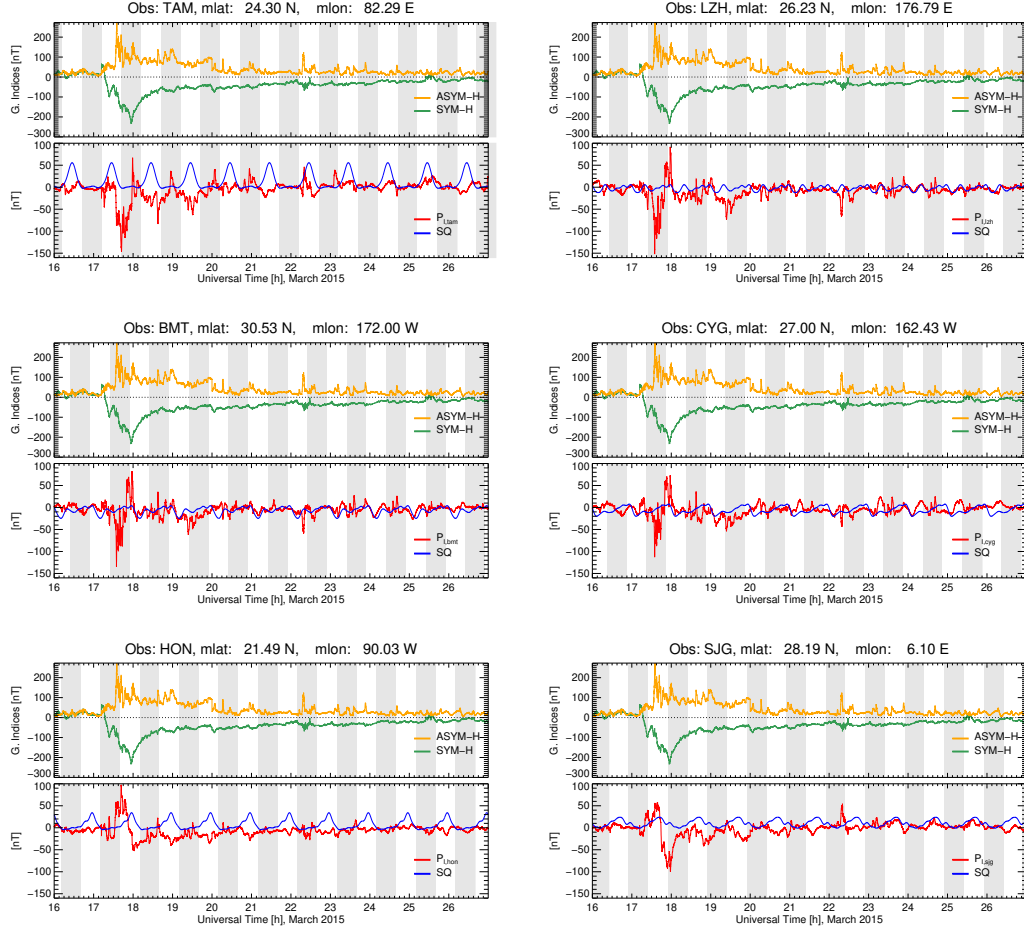


Figure 7: Geomagnetic observations for each magnetic OB. top panels show planetary geomagnetic indices SYMH (green line) and ΔH (black line). Bottom panels: Magnetic contribution of ionospheric origin (red line) and diurnal variation modeled from local quiet days (blue line) (Part 1)

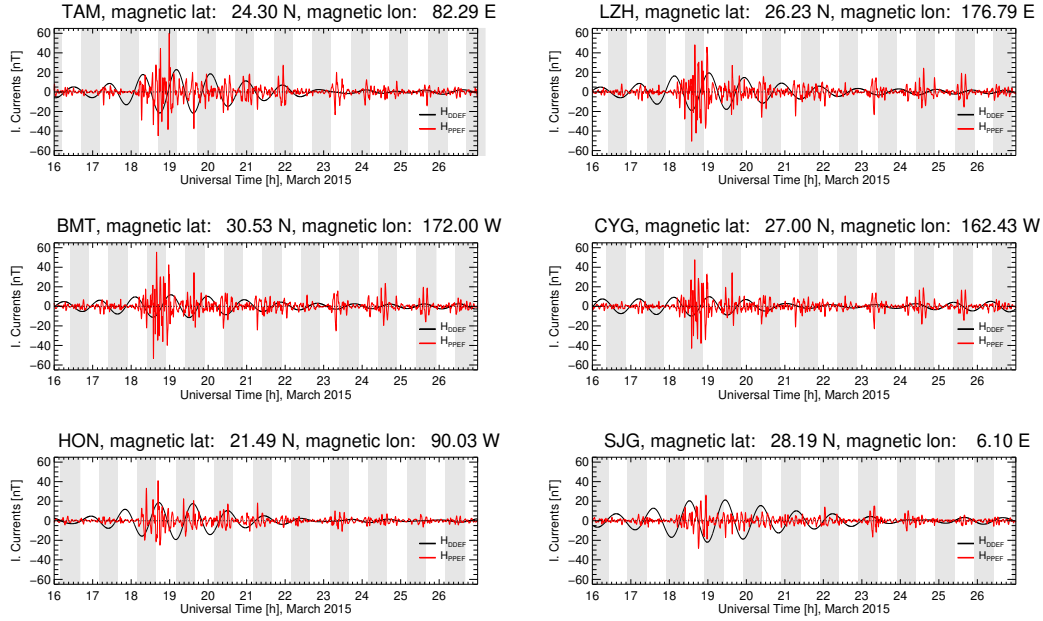


Figure 8: Geomagnetic registers for each magnetic OB. top panels show planetary geomagnetic indices SYMH (green line) and local processed magnetic registers (black line). Bottom panels: Magnetic contribution of ionospheric origin (red line) and diurnal variation modeled from local quiet days (blue line) (Part 2)

- R. J. Strangeway C.T Russell, J. G. Luhman. *SPACE PHYSICS AN INTRODUCTION*. Cambridge university press. University Pringting House, Cambridge CB2 (BS, United Kingdom, 2016. isbn: 978-1-107-09882-4.
- B. G. Fejer, M. F. Larsen, and D. T. Farley. Equatorial disturbance dynamo electric fields. *Geophysical Research Letters*, 10(7):537–540, 1983. doi:<https://doi.org/10.1029/GL010i007p00537>. <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/GL010i007p00537>. URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/GL010i007p00537>.
- J. W. Gjerloev. The SuperMAG data processing technique. *Journal of Geophysical Research (Space Physics)*, 117(A9):A09213, September 2012. doi:10.1029/2012JA017683.
- A. Guerrero, J. Palacios, M. Rodríguez-Bouza, I. Rodríguez-Bilbao, A. Aran, C. Cid, M. Herraiz, E. Saiz, G. Rodríguez-Caderot, and Y. Cerrato. Storm and Substorm Causes and Effects at Midlatitude Location for the St. Patrick’s 2013 and 2015 Events. *Journal of Geophysical Research (Space Physics)*, 122(10):9994–10, October 2017. doi:10.1002/2017JA024224.
- Veneta Guineva, Irina Despirak, and Natalia Kleimenova. Substorms manifestation at high and mid-latitudes during two large magnetic storm. *Aerospace Research in Bulgaria*, 31:27–39, January 2019. doi:10.3897/arb.v31.e03.
- Debrup Hui, D. Chakrabarty, R. Sekar, G. D. Reeves, A. Yoshikawa, and K. Shiokawa. Contribution of storm time substorms to the prompt electric field disturbances in the equatorial ionosphere. *Journal of Geophysical Research (Space Physics)*, 122(5):5568–5578, May 2017. doi:10.1002/2016JA023754.
- INTERMAGNET. International real-time magnetic observatory network, 2021. URL <https://www.intermagnet.org/index-eng.php>.
- R. K. Jaggi and Richard A. Wolf. Self-consistent calculation of the motion of a sheet of ions in the magnetosphere. *Journal of Geophysical Research*, 78:2852–2866, 1973. URL <https://api.semanticscholar.org/CorpusID:121315484>.
- M. C. Kelley, B. G. Fejer, and C. A. Gonzales. An explanation for anomalous equatorial ionospheric electric fields associated with a northward turning of the interplanetary magnetic field. *Geophysical Research Letters*, 6(4):301–304, April 1979. doi:10.1029/GL006i004p00301.

- M. V. Klimenko and V. V. Klimenko. Disturbance dynamo, prompt penetration electric field and overshielding in the Earth's ionosphere during geomagnetic storm. *Journal of Atmospheric and Solar-Terrestrial Physics*, 90:146–155, December 2012. doi:10.1016/j.jastp.2012.02.018.
- D.J Knecht and . Shuman B.M. *Handboor of Geophysics and the Space Enviromnent, chaper 4, The Geomagnetic Field.* ed by A. S. Jursa, Boston: Air Force Geophysics Laborator, 1985.
- Minh Le Huy and Christine Amory-Mazaudier. Magnetic signature of the ionospheric disturbance dynamo at equatorial latitudes: “ D_{dyn} ”. *Journal of Geophysical Research (Space Physics)*, 110(A10): A10301, October 2005. doi:10.1029/2004JA010578.
- Minh Le Huy and Christine Amory-Mazaudier. Planetary magnetic signature of the storm wind disturbance dynamo currents: D_{dyn} . *Journal of Geophysical Research (Space Physics)*, 113(A2): A02312, February 2008. doi:10.1029/2007JA012686.
- J. Y. Liu, W. H. Yang, C. H. Lin, Y. I. Chen, and I. T. Lee. A statistical study on the characteristics of ionospheric storms in the equatorial ionization anomaly region: GPS-TEC observed over Taiwan. *Journal of Geophysical Research (Space Physics)*, 118(6):3856–3865, June 2013. doi:10.1002/jgra.50366.
- Atsuhiko Nishida. Coherence of geomagnetic DP 2 fluctuations with interplanetary magnetic variations. *Journal of Geophysical Research*, 73(17):5549–5559, September 1968a. doi:10.1029/JA073i017p05549.
- Atsuhiko Nishida. Geomagnetic D_p 2 fluctuations and associated magnetospheric phenomena. *Journal of Geophysical Research*, 73(5):1795–1803, March 1968b. doi:10.1029/JA073i005p01795.
- Henry Rishbeth and Owen K. Garriott. *Introduction to ionospheric physics.* 1969.
- Christopher Torrence and Gilbert P. Compo. A practical guide to wavelet analysis. *Bulletin of the American Meteorological Society*, 79(1):61 – 78, 1998. doi:10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2. URL https://journals.ametsoc.org/view/journals/bams/79/1/1520-0477_1998_079_0061_apgtwa_2_0_co_2.xml.
- M. van de Kamp. Harmonic quiet-day curves as magnetometer baselines for ionospheric current analyses. *Geoscientific Instrumentation, Methods and Data Systems*, 2(2):289–304, November 2013. doi:10.5194/gi-2-289-2013.
- Vytenis M. Vasyliunas. Mathematical models of magnetospheric convection and its coupling to the ionosphere. In B. M. McCormac, editor, *Particles and Fields in the Magnetosphere*, pages 60–71, Dordrecht, 1970. Springer Netherlands. isbn:978-94-010-3284-1.
- Yong Wei, Biqiang Zhao, Guozhu Li, and Weixing Wan. Electric field penetration into earth's ionosphere: a brief review for 2000–2013. *Science Bulletin*, 60(8):748–761, 2015. ISSN 2095-9273. doi:<https://doi.org/10.1007/s11434-015-0749-4>. URL <https://www.sciencedirect.com/science/article/pii/S2095927316304376>.
- Chin-Chun Wu, Kan Liou, Ronald P. Lepping, Lynn Hutting, Simon Plunkett, Russ A. Howard, and Dennis Socker. The first super geomagnetic storm of solar cycle 24: “The St. Patrick’s day event (17 March 2015)”. *Earth, Planets and Space*, 68(1):151, December 2016. doi:10.1186/s40623-016-0525-y.
- W. Younas, C. Amory-Mazaudier, Majid Khan, and R. Fleury. Ionospheric and Magnetic Signatures of a Space Weather Event on 25-29 August 2018: CME and HSSWs. *Journal of Geophysical Research (Space Physics)*, 125(8):e27981, August 2020. doi:10.1029/2020JA027981.
- Waqar Younas, C. Amory-Mazaudier, Majid Khan, and M. Le Huy. Magnetic Signatures of Ionospheric Disturbance Dynamo for CME and HSSWs Generated Storms. *Space Weather*, 19(9):e02825, September 2021. doi:10.1029/2021SW002825.
- K. Z. Zaka, A. T. Koba, P. Assamoi, O. K. Obrou, V. Doumbia, K. Boka, B. J. P. Adohi, and N. M. Mene. Latitudinal profile of the ionospheric disturbance dynamo magnetic signature: comparison with the DP2 magnetic disturbance. *Annales Geophysicae*, 27(9):3523–3536, September 2009. doi:10.5194/angeo-27-3523-2009.